

$O(n)$ -size, $O(\log n)$ -depth circuits are computable by
depth-3 circuits of size $2^{O(n/\log \log n)}$
Valiant's depth reduction

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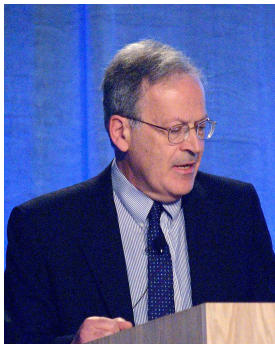
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Introduction

Leslie G. Valiant (1977): Graph-theoretic arguments in low-level complexity [7]



GRAPH-THEORETIC ARGUMENTS IN LOW-LEVEL COMPLEXITY

Leslie G. Valiant
Computer Science Department
University of Edinburgh
Edinburgh, Scotland.

1. Introduction

A major goal of complexity theory is to offer an understanding of why some specific problems are inherently more difficult to compute than others. The pursuit of this goal has two complementary fronts, the positive one of finding that algorithms, and the negative one of proving lower bounds on the inherent complexity of problems. Finding a proof of such a lower bound is equivalent to giving a property of the class of all algorithms for the problem. Because of the sheer volume of such classes, even for relatively simple problems, very little is yet understood about this and consequently the search for lower bound proofs has not with only limited success.

The poverty of our current knowledge can be illustrated by stating some major current research goals for three distinct models of computation. In each case complexity is measured in terms of n , the size of the number of inputs and outputs.

(A) Discrete problems: For some natural problem classes to be solvable in polynomial time on a multi-tape Turing machine (TM) prove that no TM exists that computes it in time $O(n^k)$. This problem is open when the TM are restricted to be oblivious [2].

(B) Discrete finite problems: For some problem computable in polynomial time on a TM show that no combinational circuit over a complete basis exists that is of size $O(n^k)$.

(C) Algebraic problems: For some natural sets of multicomplexes of constant degree over a ring show that no straight-line program consisting of the operations $+$, $\cdot$, and * , exists of size $O(n^k)$.

Known results in lower bounds are motivated by the above specifications either because they impose other restrictions on the models, or for the following reasons: For the lower bounds for natural problems have only been found for those of approach or provide exponential complexity or more $[1, 6, 11]$. For unrestricted combinatorial circuits all arguments involve counting. The only problems that have been proved to have lower complexity are those that can be solved by counting process and are of exponential complexity or more $[3, 10]$. For algebraic problems "degree arguments" have been successfully applied to natural problems, but only when the degree grows with n [5].

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Theorem (informal)

A linear-size, logarithmic-depth Boolean circuit can be converted into a subexponential-size depth-3 circuit.

Convention

Circuit can have AND, OR, and NOT gates.

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Convention

In constant-depth circuits,

- AND/OR gates have unbounded fan-ins, and
- NOT gates do not count toward depth.

Otherwise,

- AND/OR have 2 fan-ins, and
- NOT gates do count toward depth.

Convention

Logarithms are of base 2.

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Valiant's depth reduction

Theorem (Valiant's depth reduction, informal)

Let C be a linear-size logarithmic-depth Boolean circuit.

For any $\epsilon > 0$, C can be converted to a circuit s.t.

- ① *It is an OR of $2^{O(n/\log \log n)}$ CNFs; each CNF has size $2^{O(n^\epsilon)}$; and*
- ② *For every input, at most one of the CNFs outputs true.*

The new circuit is in negation-normal form.

It is of type Σ_3 , size $2^{O(n/\log \log n)}$ (if $\epsilon < 1$), and depth 3.

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Valiant's depth reduction

Theorem (Valiant's depth reduction, informal)

Let C be a linear-size logarithmic-depth Boolean circuit.

~~For any $\epsilon > 0$,~~ C can be converted to a circuit s.t.

- 1 It is an OR of $2^{O(n/\log \log n)}$ CNFs; each CNF has size $2^{O(n^{0.01})}$; and
- 2 For every input, at most one of the CNFs outputs true.

The new circuit is in negation-normal form.

It is of type Σ_3 , size $2^{O(n/\log \log n)}$ ~~(if $\epsilon < 1$)~~, and depth 3.

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Find an explicit Boolean function that cannot be computed by a circuit of linear size.

Without explicitness, resolved in the positive by a counting argument.
Best known result: $5n - o(n)$ due to Iwama and Morizumi [6].

*Find an explicit Boolean function that cannot be computed by a circuit of linear size **and logarithmic depth**.*

It suffices to show an exponential circuit lower bound for depth-3 circuits.
Best known result: $2^{\Omega(\sqrt{n})}$ due to Håstad, Jukna, and Pudlák [5].

Valiant's depth reduction has had a profound impact.
We now present its proof.

Prerequisites

Only basic complexity theory and graph theory is necessary (familiarity with Sipser)!

A circuit can be represented by a DAG, so we only consider DAGs.

DAG = directed acyclic graphs.

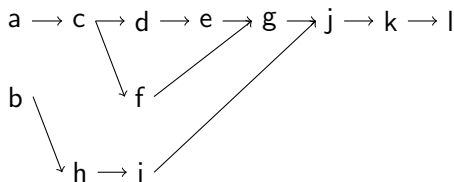


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Valiant's depth reduction is an application of a graph-theoretical result.

Definition

The **depth** of a DAG is the length of its longest path.

Definition

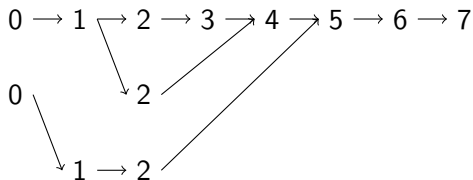
A **labeling** of a DAG with vertices V and edges E is a mapping $\ell : V \rightarrow \mathbb{N}$. A labeling ℓ is **legal** if $\ell(u) < \ell(v)$ for all edges $(u, v) \in E$.

Canonical labeling

Definition

The **canonical** labeling of a DAG with vertices V is a mapping $\ell : V \rightarrow \mathbb{N}$ where $\ell(v)$ is the length of the longest path ending at v .

Example:



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Proposition

The canonical labeling is legal.

It follows from the definitions.

Proposition

Let G be a DAG. G has a legal labeling with at most d distinct labels iff its depth is less than d .

The forward direction follows from the definitions.

The backward direction uses canonical labeling.

Removing edges: Statement

Proposition (informal)

Let $G = (V, E)$ be a DAG.

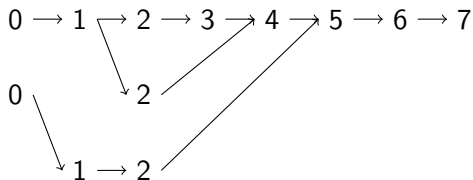
Let ℓ be its canonical labeling (in binary).

Let E_j be edges (u, v) s.t. the most significant differing bit between $\ell(u)$ and $\ell(v)$ is the j^{th} bit.

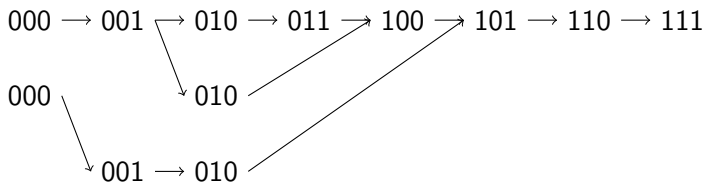
Delete E_j from G .

Deleting the j -th bit from ℓ yields a legal labeling of the new graph.

Removing edges: Example ($j = 1$)

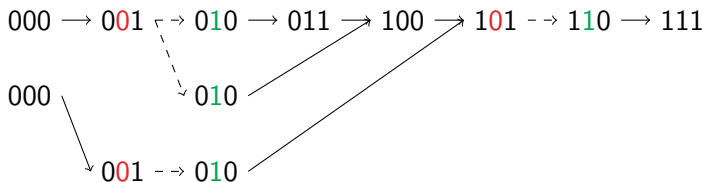


Removing edges: Example ($j = 1$)



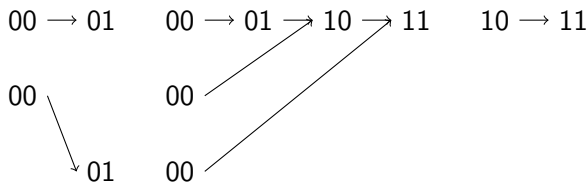
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Removing edges: Example ($j = 1$)



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Removing edges: Example ($j = 1$)



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Removing edges: Example ($j = 1$)

Case 1: the most significant differing bit position is greater than j .

$$011 \rightarrow 100$$

becomes...

$$01 \rightarrow 10$$

Removing edges: Example ($j = 1$)

Case 2: the most significant differing bit position is less than j .

$$000 \rightarrow 001$$

becomes...

$$00 \rightarrow 01$$

Removing edges: Example ($j = 1$)

Case 3: the most significant differing bit position is equal to j .

001 $\rightarrow$ 010

becomes...???

01 $\rightarrow$ 00

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Removing edges: Example ($j = 1$)

Case 3: the most significant differing bit position is equal to j .

001 $\rightarrow$ **010**

becomes...

01 00

Removing edges: Proof sketch

Proposition (informal)

Let $G = (V, E)$ be a DAG.

Let ℓ be its canonical labeling (in binary).

Let E_j be edges (u, v) s.t. the most significant differing bit between $\ell(u)$ and $\ell(v)$ is the j^{th} bit.

Delete E_j from G .

Deleting the j -th bit from ℓ yields a legal labeling of the new graph.

Proof sketch.

Let ℓ' be the new labeling for $G' = (V, E')$ where $E' = E \setminus E_j$.

It suffices to show that $\ell'(u) < \ell'(v)$ for all $(u, v) \in E'$.

AFSOC that $\ell'(u) \geq \ell'(v)$ for some $(u, v) \in E'$.

We had $\ell(u) < \ell(v)$, so deleting the j -th bit flips the inequality.

$\implies$ The j -th bit is the most significant bit at which $\ell(u)$ and $\ell(v)$ differ.

$\implies (u, v) \in E_j \implies E' = E \setminus E_j \not\ni (u, v)$, contradiction! □

Depth reduction: Proof sketch

Lemma

*Let G be a DAG with m edges and depth less than $d = 2^k$.
For any $1 \leq r \leq k$, depth of G can be reduced to less than $d/2^r$ by removing at most $(r/k) \cdot m$ edges.*

Proof sketch (continued...)

We prove this by strong induction on k . Suppose that the lemma can be applied to all DAGs of depth less than 2^{k-1} . (★)

Let V and E be the set of vertices and m edges in G .

Define j^* such that E_{j^*} is the smallest set among $E_0, E_1, \dots, E_{k-1}$.

Consider the procedure that removes all edges in E_{j^*} from E .

We selected the smallest, so $|E_{j^*}|$ is at most the average size of the sets.

$\implies$ the procedure removes $t = |E_{j^*}|$ edges where $t \leq m/k$.

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Depth reduction

Proof sketch (...continued).

Let ℓ be the canonical labeling of G .

ℓ is a legal labeling on G .

ℓ uses at most d distinct labels.

Deleting the j^* -th bit in ℓ yields a legal labeling on the new graph (G').

The new labeling ℓ' uses 1 fewer bit than ℓ .

$\implies \ell'$ uses at most $d/2$ distinct labels.

$\implies G'$ has depth less than $d/2$.

If $r = 1$, we are done.

Otherwise, use $(\star)$.

We can reduce depth to less than $(d/2)/2^{r-1} = d/2^r$ by...

...removing at most $t + ((r-1)/(k-1)) \cdot (m-t) \leq (r/k) \cdot m$ edges. $\square$

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We show that a circuit of linear size and logarithmic depth can be converted to a circuit of subexponential size and depth 3.
We begin with an intermediate result.

Intermediate result: Proof sketch

Proposition (informal)

Consider a circuit of size $O(n)$ and depth $O(\log n)$.

We can cut $O(n/\log \log n)$ wires to get a collection of subcircuits s.t. each subcircuit accepts $O(n^{0.01})$ input bits.

Proof sketch (continued...)

The circuit can be represented by a DAG G of size $c_1 n$ and depth less than $d = c_2 \log n + 1$ for some constants c_1 and $c_2 \geq 0.01$.

Let m be the number of edges in G .

Clearly, $m \leq 2c_1 n$.

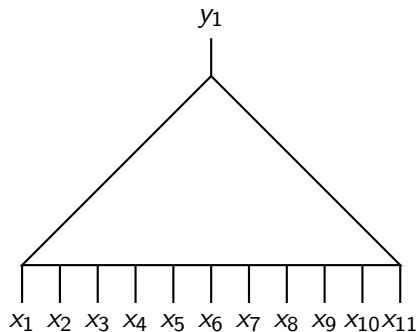
Let $r = \log(200c_2)$ and $k = \log d$.

Clearly, G has depth less than $d = 2^k$.

Clearly, $r \geq \log(200(0.01)) = 1$ and $r \ll k$.

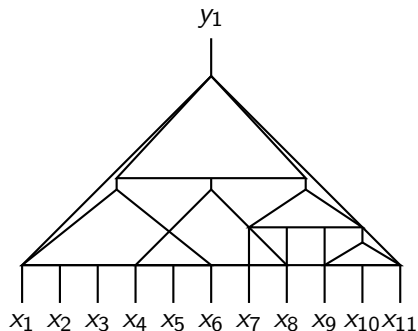
$\implies$ We can apply our lemma!

Intermediate result: example



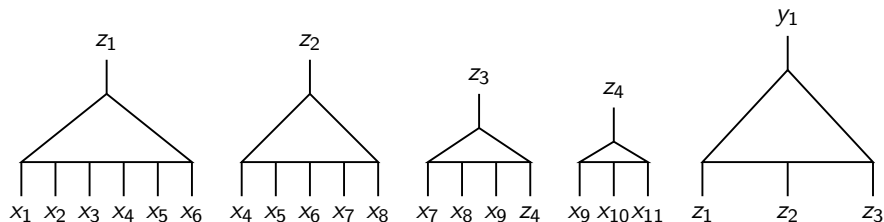
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Intermediate result: example



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Intermediate result: example



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Intermediate result: Proof sketch

Proof sketch (...continued...)

Using our lemma, the depth of G can be reduced to less than

$$\frac{d}{2^r} = \frac{2^{\log(c_2 \log n + 1)}}{2^{\log(200c_2)}} < 0.01 \log n + \frac{1}{100c_2}$$

by deleting at most

$$\left(\frac{r}{k}\right) \cdot m = \frac{rm}{k} \leq \frac{\log(200c_2) \cdot 2c_1 n}{\log(c_2 \log n + 1)} = O\left(\frac{n}{\log \log n}\right)$$

edges.

Intermediate result: Proof sketch

Proof sketch (...continued).

For each cut-wire source/original output bit, construct a subcircuit by treating it as the output bit and its ancestors as input bits or gates.

Clearly, each subcircuit has depth less than $0.01 \log n + 1/100c_2$.

Each subcircuit has at most $2^{0.01 \log n + 1/100c_2} = O(n^{0.01})$ input bits. □

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Valiant's depth reduction: Statement

Theorem (Valiant's depth reduction, informal)

Let C be a linear-size logarithmic-depth Boolean circuit.

C can be converted to a circuit s.t.

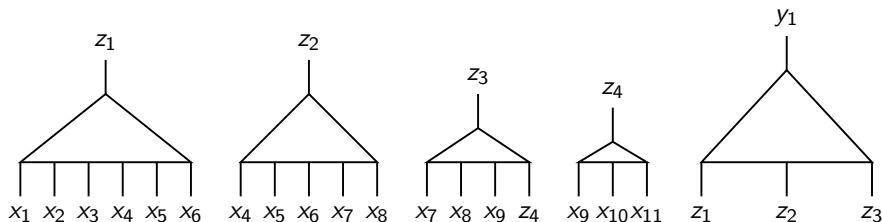
- ① *It is an OR of $2^{O(n/\log \log n)}$ CNFs; each CNF has size $2^{O(n^{0.01})}$; and*
- ② *For every input, at most one of the CNFs outputs true.*

P $\neq$ NP

Valiant's depth reduction: Proof sketch

Proof sketch (continued...)

Cut $O(n/\log \log n)$ wires to get subcircuits, each w/ $O(n^{0.01})$ input bits.

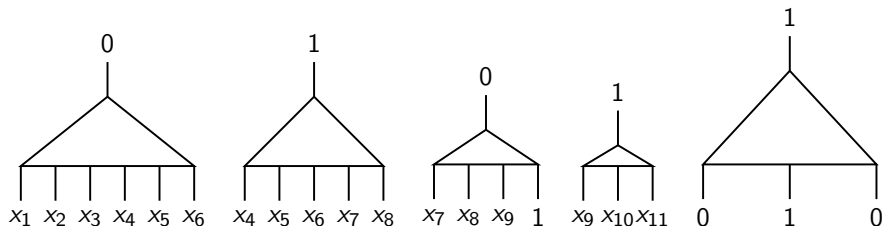


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Valiant's depth reduction: Proof sketch

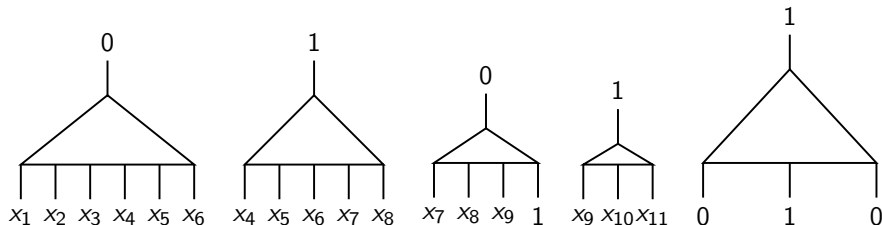
Proof sketch (...continued...)

Fix a possible Boolean assignment to the cut wires and the output bit.



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Valiant's depth reduction: Proof sketch

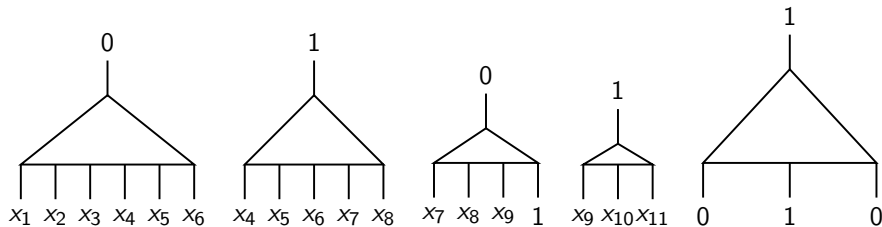


Proof sketch (...continued...)

Cut wire/output consistency check depends on $O(n^{0.01})$ original input bits.
 $\implies \exists$ equivalent CNF with at most $2^{O(n^{0.01})}$ clauses.

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Valiant's depth reduction: Proof sketch



Proof sketch (...continued...)

Need the entire cut wire/output assignment to be consistent w/ the input.
AND the CNFs for all cut wires/output bit to form a larger CNF.
The larger CNF has $(O(n/\log \log n) + 1) \cdot 2^{O(n^{0.01})} = 2^{O(n^{0.01})}$ clauses.

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Valiant's depth reduction: Proof sketch

Proof sketch (...continued...)

Do this for the $2^{O(n/\log \log n)}$ possible Boolean assignments.

We only need one assignment to work, so the larger CNFs can be OR'd.

This has size $2^{O(n/\log \log n)} \cdot 2^{O(n^{0.01})} = 2^{O(n/\log \log n)}$.

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Valiant's depth reduction: Proof sketch

Proof sketch (...continued...)

Clearly, this is OR of $2^{O(n/\log \log n)}$ CNFs, each of size $2^{O(n^{0.01})}$.

$\implies$ Property 1 satisfied!

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Valiant's depth reduction: Proof sketch

Proof sketch (...continued).

An input is consistent with exactly one cut-wire assignment.

If input is in the language, then exactly one of the larger CNFs outputs 1.
Otherwise, none does.

$\implies$ Property 2 satisfied!



Valiant's depth reduction: Remarks

After nondeterministically guessing $O(n / \log \log n)$ global bits,
Each bit consistency check only requires reading $O(n^{0.01})$ input bits.

Valiant's depth reduction: Remarks

This concludes our analysis of Valiant's depth reduction.
We now discuss its impact and applications.

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Valiant's result is merely a lower-bound reduction, not a finished lower-bound theorem.

It has nonetheless initiated a vast line of research on the study of depth-3 circuits, which cite Valiant's work as the main motivation.

Valiant's result has also been used as an ingredient to prove other notable circuit-complexity results [1].

That said, despite its enormous impact in the field, there is yet to be a superlinear general Boolean circuit lower bound.

Valiant's results have also spawned alternative interpretations.

For example, Viola [8] observed that, for a many-output function, the cut-wire values can be viewed as a small amount of preprocessed common information.

This gives rise to static data structures with small query time, whose lower bounds would imply circuit lower bounds.

Golovnev, Kulikov, and Williams [3] extended Valiant's program to show that improvements to several combinatorial objects would yield better general circuit lower bounds.

The theorem also had an outsized impact on the topic of matrix rigidity.

Matrix rigidity

Suppose that a linear map $x \mapsto Ax$ is computed by a linear-size, logarithmic-depth, bounded-fan-in linear circuit over a field.

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Matrix rigidity

After applying Valiant's depth reduction, each output depends on the $O(n/\log \log n)$ cut wire values and surviving original inputs.

Treating the cut wire values as independent inputs, the matrix A decomposes roughly as $A = L + S$, where $\text{rank}(L) = O(n/\log \log n)$ and S has $O(n^\epsilon)$ non-zeros in each row, and thus $\text{nnz}(S) = O(n^{1+\epsilon})$.

Exhibiting a matrix A that remains high-rank after changing $O(n^{1+\epsilon})$ entries $\implies$ a linear-size log-depth linear circuit lower bound.

There are also analogous results in the study of arithmetic circuits where general arithmetic circuits can be reduced to depth-3 or -4 arithmetic circuits [2, 4].

That said, the underlying mechanism is quite different:

Valiant's depth reduction uses graph-theoretical results with nondeterminism.

Arithmetic depth reduction exploits the structure of arithmetic operations.

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We examined the main result of Valiant's 1977 paper titled "Graph-theoretic arguments in low-level complexity".

Valiant proposed an easier circuit-lower bound problem, and he showed that a linear-size logarithmic-depth Boolean circuit can be converted into a subexponential-size depth-3 circuit [7].

Conclusions

We proved Valiant's depth reduction result using his graph-theoretical proof ingredients.

We discussed the theorem's impact and its applications beyond simply proving circuit lower bounds.

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Valiant's depth reduction leaves us with a striking challenge: Prove that shallow circuits need exponential size to show that deeper circuits need more than linear size.

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Acknowledgements

The proof structure was inspired by Problem 1 in Homework 1 and Problem 3 in Homework 2 of the fall 2017 course 15-855: Graduate Computational Complexity Theory taught by Professor Ryan O'Donnell from Carnegie Mellon University.

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References



Amir Abboud and Aviad Rubinfeld.

Fast and Deterministic Constant Factor Approximation Algorithms for LCS Imply New Circuit Lower Bounds.
In Proceedings of the Innovations in Theoretical Computer Science Conference (ITCS), 2018.



Manindra Agrawal and Vishwanathan Vinay.

Arithmetic circuits: A chasm at depth four.
In Proceedings of the IEEE Symposium on Foundations of Computer Science (FOCS), 2008.



Alexander Golovnev, Alexander S. Kulikov, and Ryan R. Williams.

Circuit depth reductions.
Technical Report 192, Electronic Colloquium on Computational Complexity (ECCC), 2019.



Ankit Gupta, Pritish Kamath, Neeraj Kayal, and Ramprasad Saptharishi.

Arithmetic circuits: A chasm at depth 3.
SIAM Journal on Computing, 45(3):1064–1079, 2016.



Johan Håstad, Stasys Jukna, and Pavel Pudlák.

Top-down lower bounds for depth 3 circuits.
In Proceedings of the IEEE Annual Foundations of Computer Science (FOCS), 1993.



Kazuo Iwama and Hiroki Morizumi.

An explicit lower bound of $5n - o(n)$ for boolean circuits.
In Proceedings of the International Symposium on Mathematical Foundations of Computer Science (MFCS), 2002.



Leslie G. Valiant.

Graph-theoretic arguments in low-level complexity.
In Proceedings of the International Symposium on Mathematical Foundations of Computer Science (MFCS), 1977.



Emanuele Viola.

Lower bounds for data structures with space close to maximum imply circuit lower bounds.
Technical Report 186, Electronic Colloquium on Computational Complexity (ECCC), 2018.

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